

Offline Setup Guide

What Needs Internet, What Doesn't, and What Each
Package Does

A one-time checklist so you can learn Python and work on the
capstone project without needing a live connection.

The Short Version

You need internet **once**, to install things. After that, you can write Python, run the capstone project, and run your tests completely offline — because the project talks to a fake server running on your own machine, not the real internet.

1. What Needs Internet vs. What Doesn't

Needs internet (one-time only)

- Installing Python itself
- Installing the packages this project needs (`pip install ...`)

Needs internet (only if you choose to use it)

- Looking things up, docs, this chat
- Pointing the project at a *real* API or website instead of the mock server

Works completely offline

- Writing and running plain Python (variables, loops, dictionaries, functions, classes, files, exceptions — basically all of *Python Crash Course*)
 - Running `pytest` on your own test files
 - Running `cProfile`
 - Reading any PDF you've already downloaded
 - Running `mock_server.py` and `main.py` together — they only talk to `127.0.0.1` (your own machine), never the real internet
 - Everything in the capstone project, once packages are installed
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2. What Each Package Actually Is

These are **third-party libraries** — not built into Python, so you install them once with `pip`, and then they live locally on your machine.

Package	What it does	Where it's used
flask	A small web framework — lets a Python script pretend to be a real website/API	Powers <code>mock_server.py</code> , which fakes the API and review site so you can practice without needing real internet endpoints
requests	Lets Python make HTTP calls (visit a URL from code) and read the response	Fetching product data from the API
beautifulsoup4 (imported as <code>bs4</code>)	Parses HTML so you can pull out specific pieces of data	Extracting the review score from inside a webpage's HTML
aiohttp	Like <code>requests</code> , but built for <code>asyncio</code> (the async/concurrent style)	The async version of fetching data
pytest	Finds and runs your test files, reports pass/fail	Running <code>test_project.py</code>

3. One-Time Offline Setup Checklist

Do this once while you have wifi or mobile data. After this, you're set.

Step 1 — Install Python (needs internet, once)

Windows: - Download from python.org/downloads - During install, check **"Add Python to PATH"** - Verify: open Command Prompt and run `python --version`

Ubuntu:

```
sudo apt update
sudo apt install python3 python3-pip python3-venv
python3 --version
```

Step 2 — Create your project folder

```
mkdir data_aggregator
cd data_aggregator
```

Step 3 — Create a virtual environment

A virtual environment is an isolated copy of Python just for this project, so installed packages don't clash with other projects.

Windows:

```
python -m venv venv
venv\Scripts\activate
```

Ubuntu:

```
python3 -m venv venv
source venv/bin/activate
```

You'll know it worked because your terminal prompt now starts with `(venv)`.

Step 4 — Install all packages in one shot (needs internet, once)

```
pip install flask requests beautifulsoup4 aiohttp pytest
```

Step 5 — Confirm it worked

```
python -c "import flask, requests, bs4, aiohttp, pytest; print('all good')"
```

If you see `all good` with no errors, you're fully set up.

4. Your Normal Offline Workflow From Here

Once setup is done, you never need internet again for this project:

1. Activate your virtual environment (`venv\Scripts\activate` or `source venv/bin/activate`)
2. In one terminal: `python mock_server.py` (leave it running)
3. In a second terminal (venv activated again): `python main.py --mode threads`
4. Run tests anytime: `pytest test_project.py -v`

All of this runs locally. The "API" and "website" the project talks to are just your own laptop pretending to be a server at `127.0.0.1` — no real network traffic leaves your machine.

5. A Few Other Things Worth Knowing

- **You only need to reinstall packages if you delete the `venv` folder or move to a new computer.** Otherwise, the install from Step 4 sticks around indefinitely.
- **If you switch computers**, you'll need internet again briefly to redo Step 4 (or Steps 1–4 if Python isn't installed there either).
- **Editors like VS Code work fine offline** once installed, though the initial download and any extensions need internet the first time.
- **Don't confuse "no internet" with "no packages."** A common beginner trip-up is assuming Python has these libraries built in — it doesn't. If you ever see `ModuleNotFoundError: No module named 'flask'` (or `requests/bs4/aiohttp/pytest`), it just means you're not inside the virtual environment where you installed it, or you skipped Step 4 on this machine.
- **Battery/data-saving tip:** since your timetable showed weekend sessions and a phone likely doubling as a hotspot, it's worth doing the full setup once at home or somewhere with reliable wifi, rather than relying on mobile data during a live session.